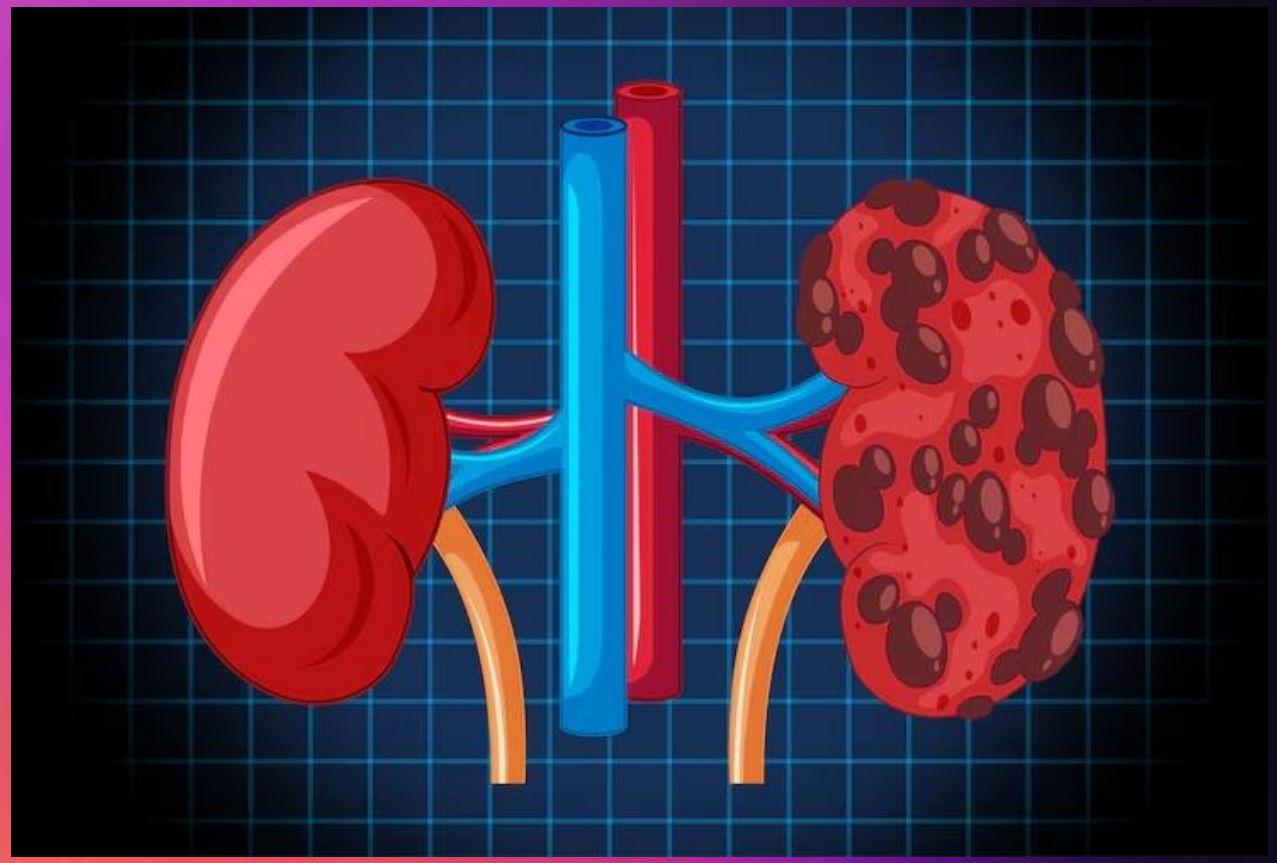


Kidney Disease

Rin Hwang

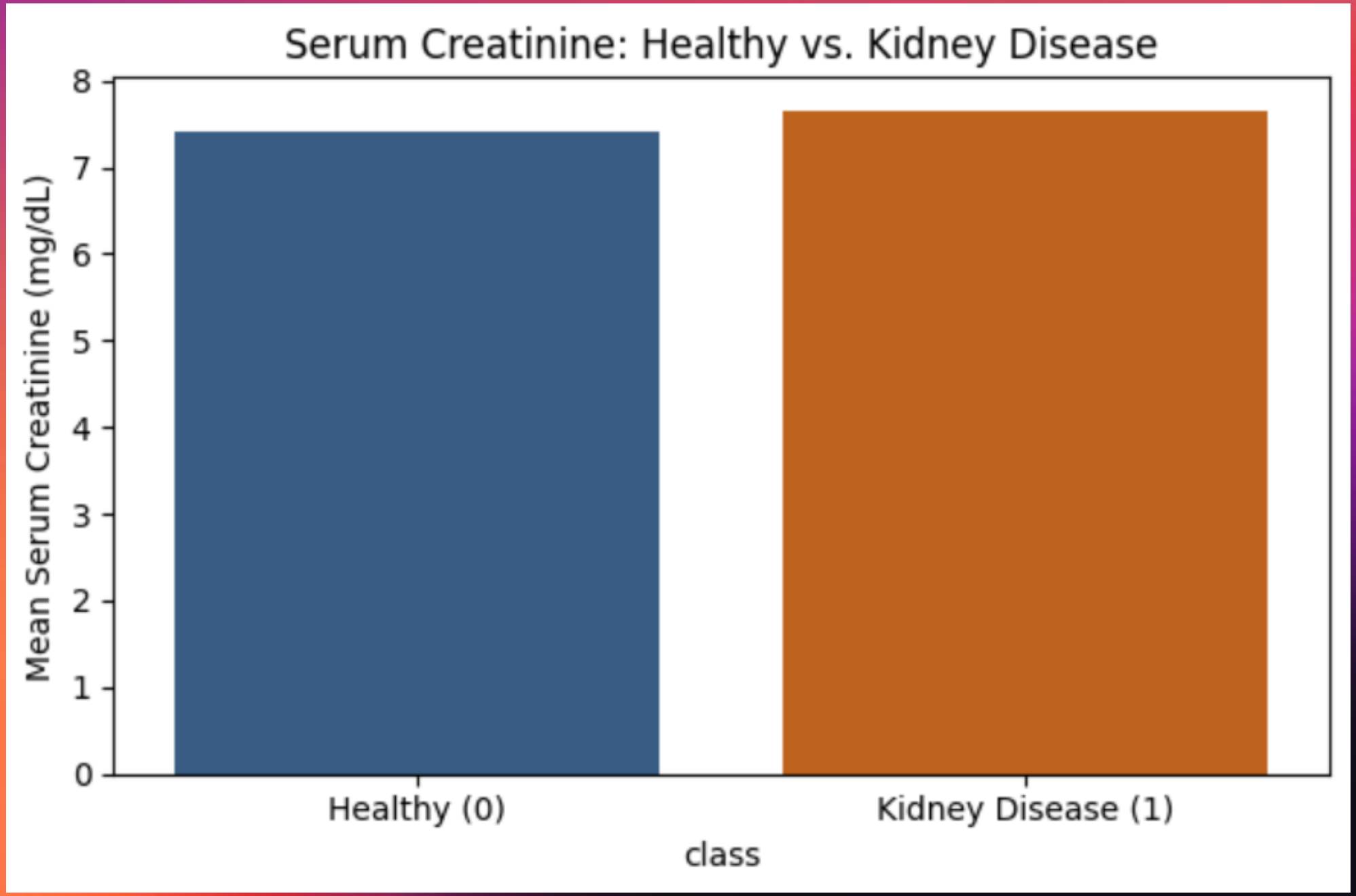
DATA 201

The Problem

- Can we catch kidney disease before it is too late?
- What we are predicting: Whether a patient has kidney disease [1] or is healthy [0] using lab work results.
- Why it matters:
  - Kidney disease develops quietly
  - Patients often feel completely normal until major damage is already done.
  - Existing damage is irreversible
- Early detection gives doctors time for lifestyle and medication changes that can prevent permanent kidney failure or need for dialysis.
- The goal: Build a reliable screening classifier that flags at-risk patients early and keeps missed cases as close to zero as possible.

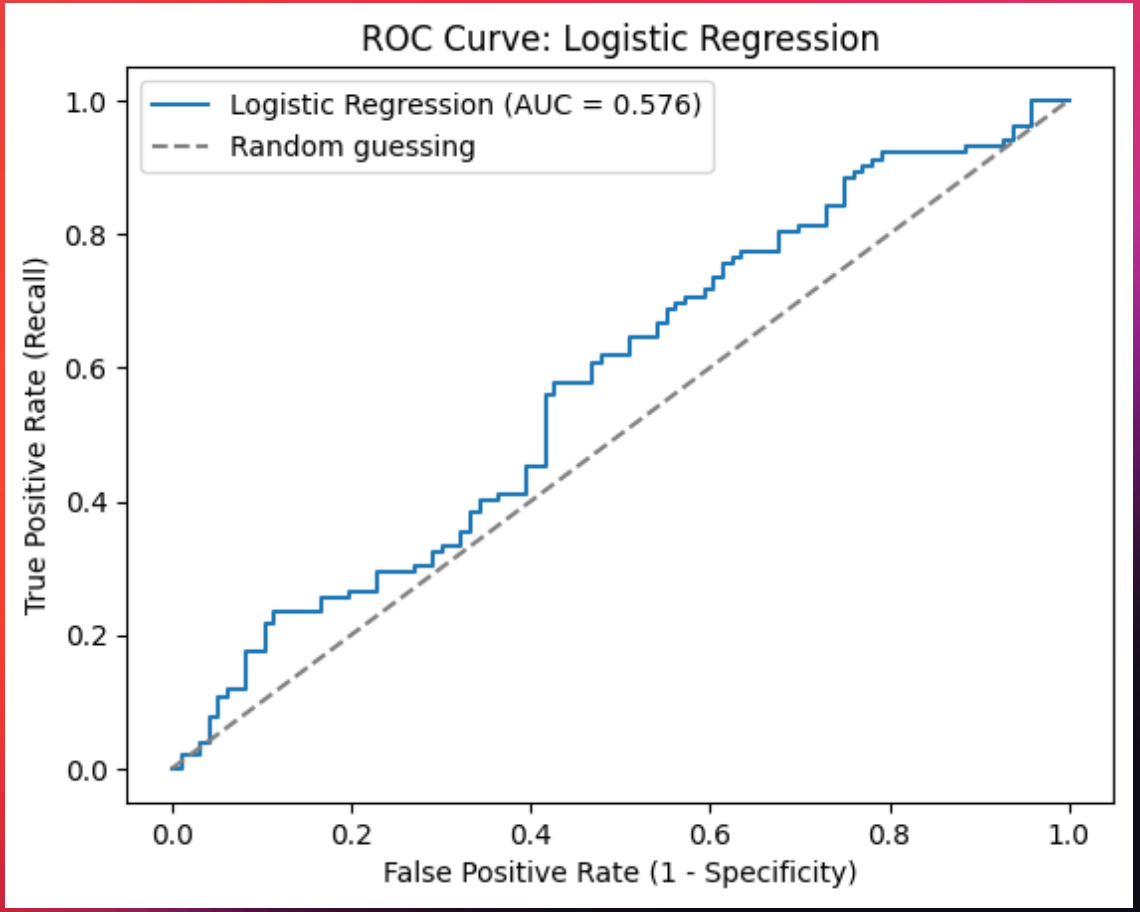
The Data

- From “BD-KDD: A real-world clinical dataset for kidney disease diagnosis and healthy classification,” June 2026
- 988 patient records with 24 predictors (blood glucose, blood pressure, age, etc.)
- Balanced: 51% kidney disease vs. 49% healthy


Serum Creatinine: Healthy vs. Kidney Disease
Mean Serum Creatinine (mg/dL)
0
1
2
3
4
5
6
7
8
Healthy (0)
Kidney Disease (1)
class

Models

Model	Accuracy	Precision	Recall	Specificity	F1	ROC-AUC
Logistic	0.571	0.579	0.608	0.531	0.593	0.576
LDA	0.566	0.574	0.608	0.521	0.590	0.576
QDA	0.525	0.543	0.500	0.552	0.520	0.520
KNN	0.556	0.561	0.627	0.479	0.593	0.558



- All four models score between 52% and 57% accuracy, barely beating a 50/50 guess.
- The low AUC shows that altering thresholds cannot fix the underlying lack of separation.

The Decision

- Logistic regression performed best out of all of them
- Top accuracy (57.1%), but not good enough
- Do not deploy in a real clinic
- Best model still missed 40 out of 102 sick test patients
- False positives = healthy person is labeled as sick
- False negatives = sick person is labeled as healthy
  - Far more dangerous because a patient with active kidney disease is sent home without early treatment
- Limitation: Data is only from one diagnostic center
  - Healthy patients in the dataset have creatinine levels up to 15 mg/dL, which normally signals serious renal failure